

ST5209/X Assignment 3

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Set up

1. Make sure you have the following installed on your system: $\text{\LaTeX}$, R4.2.2+, RStudio 2023.12+, and Quarto 1.3.450+.
2. Pull changes from the course [repo](#).
3. Create a separate folder in the root directory of the repo, label it with your name, e.g. `yanshuo-assignments`
4. Copy the `assignment1.qmd` file over to this directory.
5. Modify the duplicated document with your solutions, writing all R code as code chunks.
6. When running code, make sure your working directory is set to be the folder with your assignment `.qmd` file, e.g. `yanshuo-assignments`. This is to ensure that all file paths are valid.¹

Submission

1. Render the document to get a `.pdf` printout.
2. Submit both the `.qmd` and `.pdf` files to Canvas.

Warning: package 'ggplot2' was built under R version 4.5.2

Warning: package 'readr' was built under R version 4.5.2

1. Holt-Winters, residuals, and forecast accuracy

Consider the antidiabetic drug sales time series which can be loaded using the following code snippet.

¹You may view and set the working directory using `getwd()` and `setwd()`.

```
diabetes <- read_rds("../_data/cleaned/diabetes.rds") |>
  select(TotalC)
```

a. Fit the following exponential smoothing models on the entire time series:

- Holt-Winters with multiplicative noise and seasonality,
- Holt-Winters with a log transformation, with additive noise and seasonality,
- Holt-Winters with multiplicative noise and seasonality, and damping.

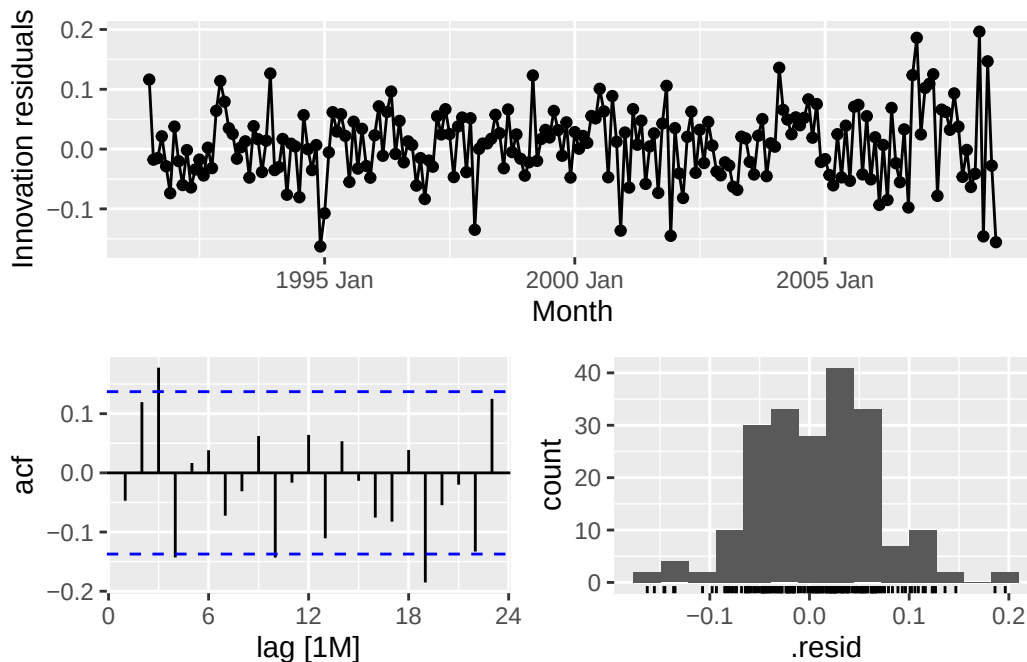
```
diabetes<-as_tsibble(diabetes)
dia_fit<-diabetes|>
  model(
    mult=ETS(TotalC~error("M")+trend("A")+season("M")),
    log=ETS(log(TotalC)~error("A")+trend("A")+season("A")),
    mult_damp=ETS(TotalC~error("M")+trend("Ad")+season("M"))
  )
```

b. Make ACF plots for the innovation residuals of these three models. Does this suggest that the residuals are drawn from a white noise model?

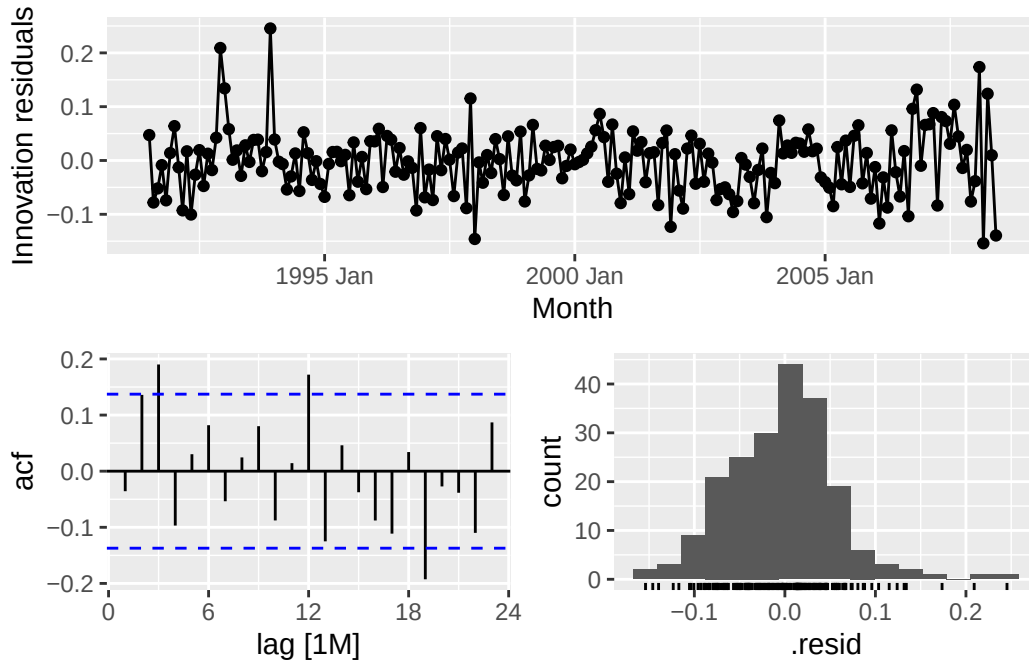
```
gg_tsresiduals(select(dia_fit,mult))
```

Warning: `gg\_tsresiduals()` was deprecated in feasts 0.4.2.

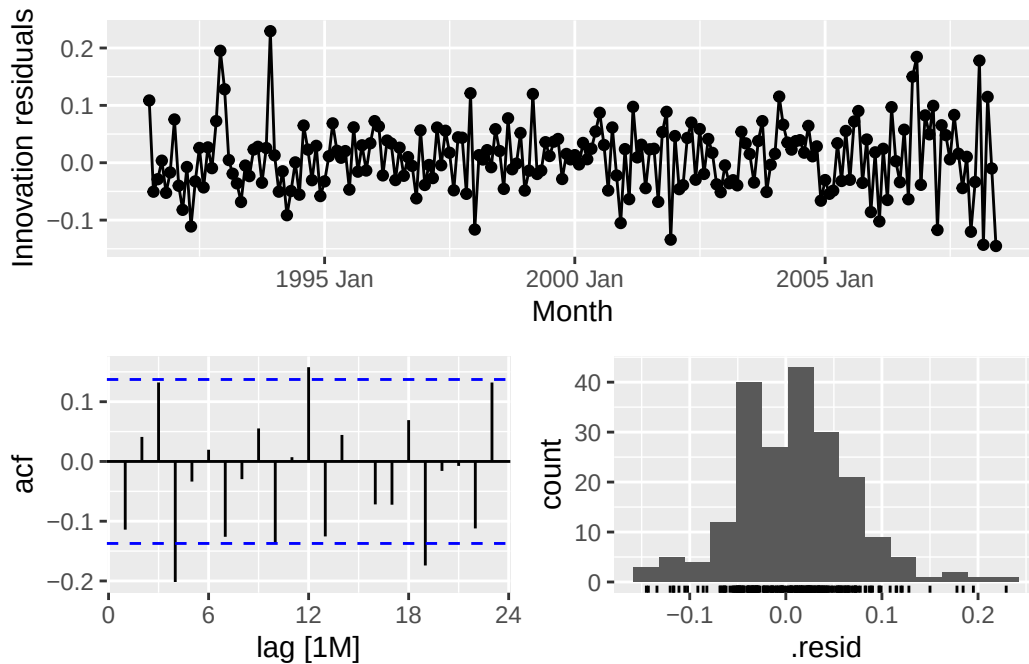
i Please use `ggtime::gg\_tsresiduals()` instead.



```
gg_tsresiduals(select(dia_fit,log))
```



```
gg_tsresiduals(select(dia_fit,mult_damp))
```



The ACF plots for residuals of each model shows significant spikes (autocorrelations),

suggesting the residuals are not consistent with white noise.

- c. Calculate the p-value from a Ljung-Box test on the residuals with lag $h = 8$. Does this suggest that the residuals are drawn from a white noise model? What does this mean about the fitted model?

```
dia_fit|>
  augment()|>
  features(.innov,ljung_box,lag=8)
```

```
# A tibble: 3 x 3
  .model    lb_stat lb_pvalue
  <chr>      <dbl>   <dbl>
1 log         16.0    0.0424
2 mult        16.0    0.0420
3 mult_damp   19.2    0.0141
```

Since all the p-values are lower than 0.05, we reject the null hypothesis that the residuals are white noise. This means these models don't have a good fit to the data.

- d. Perform time series cross-validation for the three methods, using `.init = 50` and `.step = 10`, and with the forecast horizon $h = 4$. Which method has the best RMSSE? How many data points is the error averaged over in total?

```
dia_cv<-diabetes|>
  stretch_tsibble(.step = 10,.init = 50)|>
  model(
    mult=ETS(TotalC~error("M")+trend("A")+season("M")),
    log=ETS(log(TotalC)~error("A")+trend("A")+season("A")),
    mult_damp=ETS(TotalC~error("M")+trend("Ad")+season("M"))
  )

dia_forc<-dia_cv|>
  forecast(h=4)

dia_forc|>
  accuracy(diabetes)|>
  select(.model,RMSSE)|>
  arrange(RMSSE)
```

```
# A tibble: 3 x 2
  .model    RMSSE
  <chr>      <dbl>
1 log         0.655
2 mult_damp   0.659
```

```
3 mult      0.683
```

We can find the log with additive method has the best RMSSE.

```
n<-nrow(diabetes)
steps<-ceiling((n-50)/10)
h<-4
print(steps*h)
```

```
[1] 64
```

So the error is averaged over 64 data points.

2. ETS parameters, prediction intervals

The dataset `hh_budget` contains annual indicators of household budgets for a few countries.

- a. Fit ETS on the time series comprising savings as a portion of household budgets in Canada. Do not specify what type of ETS model to fit, instead allowing the function to perform automatic model selection.

```
canada_savings<-hh_budget|>
  filter(Country=="Canada")|>
  select(Year,Savings)|>
  as_tsibble(index=Year)
fit_savings<-canada_savings|>
  model(ETS(Savings))
```

- b. Which ETS model was selected?

```
fit_savings
```

```
# A mable: 1 x 1
# ETS(Savings)
# <model>
1 <ETS(A,N,N)>
```

ETS(A,N,N) was selected, which is simple exponential smoothing with only additive error.

- c. What are the fitted parameters of the model?

```
fit_savings|>tidy()
```

```
# A tibble: 2 x 3
  .model      term estimate
  <chr>      <chr>   <dbl>
1 ETS(Savings) alpha    1.000
2 ETS(Savings) l[0]     9.31
```

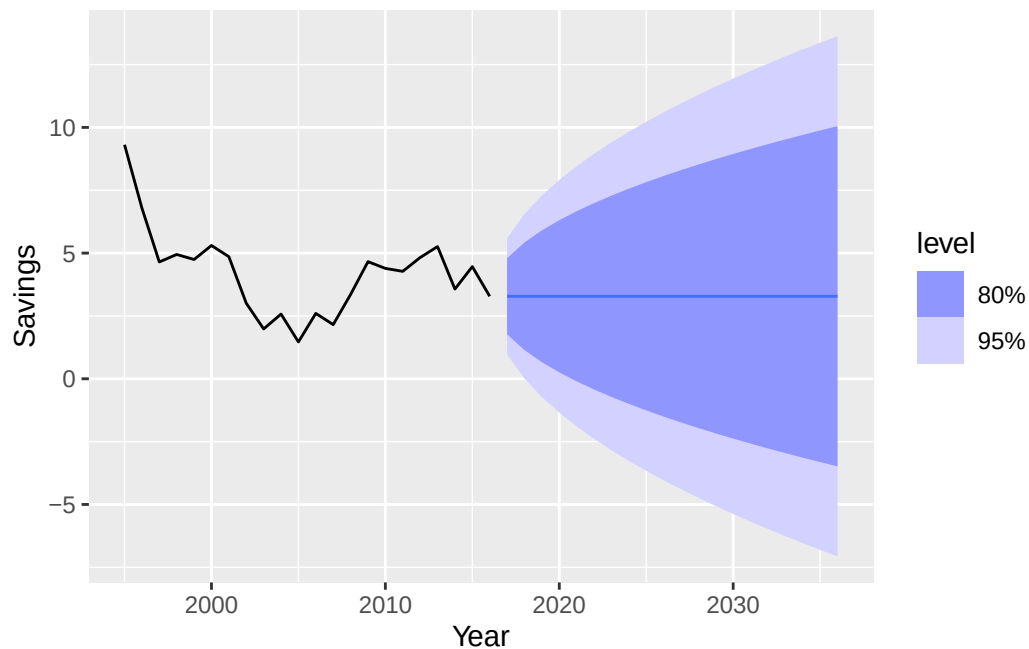
- d. Based on the fitted parameters, what simple forecasting method is this similar to?

Since $\alpha \approx 1$, this is similar to naive method.

- e. Based on your answer to d), how does the width of the prediction interval change as a function of the forecast horizon h ? Make a plot to verify this.

The prediction interval will widen as the forecast horizon increases, which means the forecast variance increases with h . This can be also be interpreted through the variance of naive method, which is $Var(e_h) = h\sigma^2$.

```
fc<-fit_savings|>
  forecast(h=20)
autoplot(fc,canada_savings)
```



The plot supports our conclusion.

3. Moving averages and differences

Consider the linear trend model

$$X_t = \beta_0 + \beta_1 t + W_t.$$

Define a time series (Y_t) by taking a moving average of (X_t) with a symmetric window of size 7. Define another times series (Z_t) by taking a difference of (X_t) .

- a. What is the mean function for (Y_t) ? What is the ACVF for (Y_t) ?

$$Y_t = \frac{1}{7} \sum_{k=-3}^3 X_{t+k} Z_t = X_t - X_{t-1}$$

The mean function is

$$\mu_{Y_t} = E[Y_t] = \frac{1}{7} \sum_{k=-3}^3 E[X_{t+k}] = \frac{1}{7} \sum_{k=-3}^3 (\beta_0 + \beta_1(t+k)) = \beta_0 + \beta_1 t + \frac{\beta_1}{7} \sum_{k=-3}^3 k = \beta_0 + \beta_1 t$$

The ACVF is

$$\gamma_Y(t, t+h) = \text{Cov}(Y_t, Y_{t+h}) = \frac{1}{49} \sum_{j=-3}^3 \sum_{k=-3}^3 \text{Cov}(X_{t+j}, X_{t+h+k}) = \frac{\sigma^2}{49} \cdot |\{(j, k) : j-k = h, j, k \in \{-3, \dots, 3\}\}|$$

The number of valid pairs is $m(h) = \max(7 - |h|, 0)$, so:

$$\gamma_Y(h) = \begin{cases} \frac{\sigma^2}{49}(7 - |h|) & |h| \leq 6 \\ 0 & |h| > 6 \end{cases}$$

- b. What is the mean function for (Z_t) ? What is its ACVF?

The mean function is

$$\mu_Z(t) = E[Z_t] = E[X_t - X_{t-1}] = (\beta_0 + \beta_1 t) - (\beta_0 + \beta_1(t-1)) = \beta_1$$

The ACVF is

$$\gamma_Z(h) = \text{Cov}(Z_t, Z_{t+h}) = \text{Cov}(W_t - W_{t-1}, W_{t+h} - W_{t+h-1})$$

Using the white noise property $\text{Cov}(W_s, W_r) = \sigma^2 \cdot \mathbf{1}(s = r)$:

$$\gamma_Z(h) = \begin{cases} 2\sigma^2 & h = 0 \\ -\sigma^2 & |h| = 1 \\ 0 & |h| \geq 2 \end{cases}$$

c. What is the CCF of (Y_t) and (Z_t) ?

$$\gamma_{YZ}(h) = \text{Cov}(Y_t, Z_{t+h}) = \text{Cov}\left(\frac{1}{7} \sum_{j=-3}^3 X_{t+j}, X_{t+h} - X_{t+h-1}\right) = \frac{\sigma^2}{7} \sum_{j=-3}^3 [\mathbf{1}(j = h) - \mathbf{1}(j = h - 1)]$$

Evaluating over $j \in \{-3, \dots, 3\}$:

$$\gamma_{YZ}(h) = \begin{cases} \frac{\sigma^2}{7} & h = -3 \\ -\frac{\sigma^2}{7} & h = 4 \\ 0 & \text{otherwise} \end{cases}$$

d. Are (Y_t) and (Z_t) jointly stationary?

No. Since the mean function for (Y_t) is time dependent, (Y_t) is not stationary and (Y_t) and (Z_t) are not jointly stationary.

4. Sample vs population ACF

Consider the signal plus noise model

$$X_t = \sin(2\pi t/5) + W_t.$$

a. What is the ACF of (X_t) ?

The mean function is:

$$\mu_X(t) = E[X_t] = \sin(2\pi t/5) + E[W_t] = \sin(2\pi t/5)$$

The ACVF is:

$$\gamma_X(t, t+h) = \text{Cov}(X_t, X_{t+h}) = \text{Cov}(\sin(2\pi t/5) + W_t, \sin(2\pi(t+h)/5) + W_{t+h}) = \text{Cov}(W_t, W_{t+h}) = \sigma^2 \mathbf{1}(t = t+h)$$

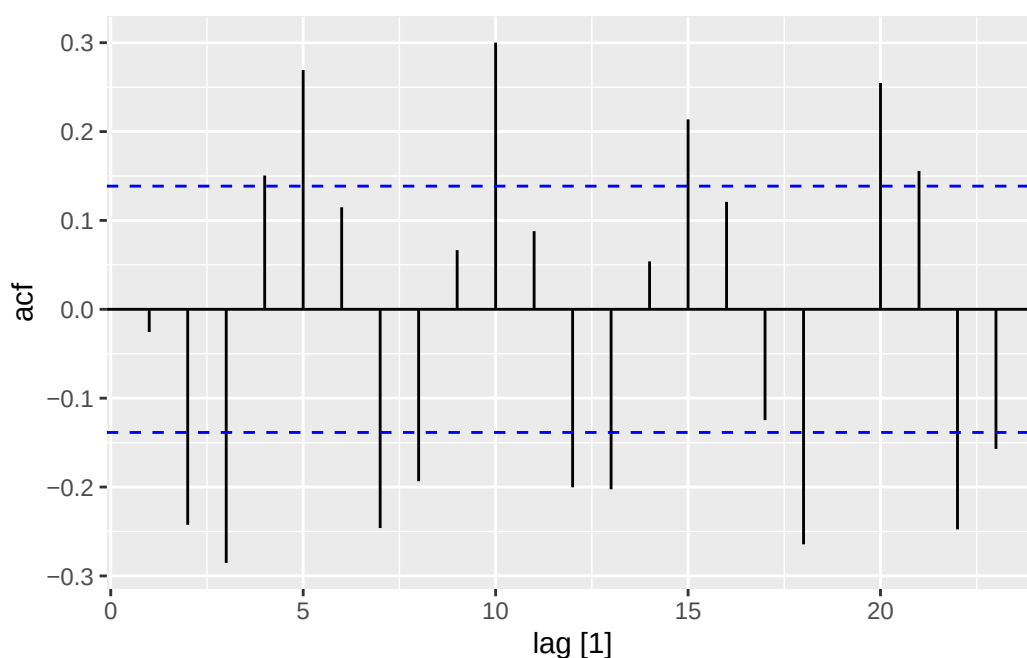
since the deterministic signal $\sin(2\pi t/5)$ contributes nothing to the covariance.

Therefore the ACF is

$$\rho_X(h) = \frac{\gamma_X(h)}{\gamma_X(0)} = \mathbf{1}(h = 0) = \begin{cases} 1 & h = 0 \\ 0 & h \neq 0 \end{cases}$$

- b. Simulate a time series $X_1, X_2, \dots, X_{200}$ from this model and plot its sample ACF.

```
set.seed(5209)
t <- 1:200
X <- sin(2 * pi * t / 5) + rnorm(200)
X_ts <- as_tsibble(tibble(t = t, X = X), index = t)
X_ts |> ACF(X) |> autoplot()
```



- c. Why does the sample ACF not look like the population ACF function?

The population ACF is $\rho_X(h) = \mathbf{1}(h = 0)$, which equals 1 at lag 0 and 0 elsewhere, resembling white noise. However, the sample ACF exhibits a sinusoidal pattern with period 5.

This discrepancy arises because (X_t) is not weakly stationary: its mean function $\mu_X(t) = \sin(2\pi t/5)$ is time-varying. The sample ACF estimator is:

$$\hat{\rho}(h) = \frac{\sum_{t=1}^{T-h} (X_t - \bar{X})(X_{t+h} - \bar{X})}{\sum_{t=1}^T (X_t - \bar{X})^2}$$

This estimator implicitly assumes stationarity by using a single grand mean $\bar{X}$ to centre the series. When the true mean is time-varying, $\bar{X}$ is a poor estimate of $\mu_X(t)$ at each time point, and the residuals $(X_t - \bar{X})$ retain the sinusoidal structure of the signal, causing the sample ACF to reflect the periodicity of $\sin(2\pi t/5)$ rather than the true population ACF.

d. Why does the asymptotic normality theorem for the ACF not apply?

The asymptotic normality does not apply because (X_t) is not weakly stationary, since its mean function $\mu_X(t) = \sin(2\pi t/5)$ depends on t .

5. Gaussian processes

Consider a random vector (X_1, X_2, X_3, X_4) that has the joint Gaussian distribution

$$\begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix} \sim N \left(\begin{bmatrix} -0.5 \\ 0 \\ 0.5 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 & 1 & 1 & 1 \\ 1 & 3 & 1 & 1 \\ 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 3 \end{bmatrix} \right)$$

```
A <- matrix(c(3, 1, 1, 1,
              1, 3, 1, 1,
              1, 1, 3, 2,
              1, 1, 2, 3),
            nrow = 4, ncol = 4, byrow = TRUE)
```

Partition the random vector as $\mathbf{X} = (\mathbf{X}_A^\top, X_4)^\top$ where $\mathbf{X}_A = (X_1, X_2, X_3)^\top$. The joint distribution gives:

$$\mu_A = \begin{bmatrix} -0.5 \\ 0 \\ 0.5 \end{bmatrix}, \quad \mu_4 = 2, \quad \Sigma_{AA} = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}, \quad \Sigma_{A4} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \quad \Sigma_{44} = 3$$

The conditional distribution formula for jointly Gaussian random vectors gives:

$$X_4 \mid \mathbf{X}_A = \mathbf{a} \sim N(\mu_{4|A}, \sigma_{4|A}^2)$$

where:

$$\mu_{4|A} = \mu_4 + \Sigma_{4A}\Sigma_{AA}^{-1}(\mathbf{a} - \mu_A)\sigma_{4|A}^2 = \Sigma_{44} - \Sigma_{4A}\Sigma_{AA}^{-1}\Sigma_{A4}$$

- a. What is the marginal variance of X_4 ?

The marginal variance of X_4 is the (4,4) entry of the covariance matrix:

$$\text{Var}(X_4) = \Sigma_{44} = 3$$

- b. What is the conditional variance of X_4 , conditioned on the observations $X_1 = 1, X_2 = 1, X_3 = 1$. Does it depend on these particular values? *Hint: The following code snippet creates matrix in R. You may use `solve()` to find the inverse of a matrix.*

```
Sigma_AA <- A[1:3, 1:3]
Sigma_A4 <- A[1:3, 4]
Sigma_4A <- A[4, 1:3]
Sigma_44 <- A[4, 4]
cond_var <- Sigma_44 - Sigma_4A %*% solve(Sigma_AA) %*% Sigma_A4
cat("Conditional variance of X_4:", cond_var, "\n")
```

Conditional variance of X_4: 1.6

- c. What is the conditional mean of X_4 , conditioned on the observations $X_1 = 1, X_2 = 1, X_3 = 1$?

```
mu_A <- c(-0.5, 0, 0.5)
mu_4 <- 2
a <- c(1, 1, 1)
cond_mean <- mu_4 + t(Sigma_A4) %*% solve(Sigma_AA) %*% (a - mu_A)
cat("Conditional mean of X4:", cond_mean, "\n")
```

Conditional mean of X4: 2.55

- d. Write a level 95% prediction interval for X_4 given these observations.

```
qnorm(0.975)
```

```
[1] 1.959964
```

```
lower <- cond_mean - 1.96 * sqrt(cond_var)
upper <- cond_mean + 1.96 * sqrt(cond_var)
cat("95% prediction interval: [", lower, ",", upper, "]\n")
```

95% prediction interval: [0.07077431 , 5.029226]